

Typhoon-Induced Turbine Failure at Wenchang Wind Power Plant

1. PROJECT OVERVIEW

1.1 Development History

1.1.1 Phase I

On February 28, 2008, the construction on Phase I of the Huaneng Wenchang wind power project (Wenchang Plant) in southern China's Hainan province was commenced (Nanhai Net, 2008). The project is located on the southeastern coast of Wenchang City in eastern Hainan Province, within a semi-hilly coastal terrain, see Figure 1. Hainan lies within a typhoon-prone region. The province is affected by 8 to 9 typhoons annually on average, with the primary typhoon season typically occurring between July and September (Haikou Meteorological Bureau, 2024). The site lies approximately 70 km from downtown Haikou and 75 km from Wenchang City, as shown in Figure 2.



Figure 1: Wind turbines along the coast of Mulan Bay, Wenchang City, in South China's Hainan Province, on February 24, 2023 (image from: Global Times, 2023).

The Wenchang Plant is the largest windfarm in Hainan and is also considered an important part of the Eleventh Five-Year Plan for the province (Zhang et al., 2010). The project has a total planned capacity of 120 megawatts (MW) with an estimated cost of \$190 million. Phase I was approved with a capacity of 49.5 MW, consisting of 33 Sinovel SL-1500 wind turbines (1.5 MW each) arranged along a 7.66 km stretch of coastline. These turbines were designed with a service life of 20 years (Nanhai Net, 2008).

The project owner and developer was Huaneng Hainan Power Co., Ltd. (Huaneng), while the wind turbine manufacturer was Sinovel Wind Group Co., Ltd. (Sinovel). The installation of the first wind turbine was successfully completed on August 16. On December 21, the first batch

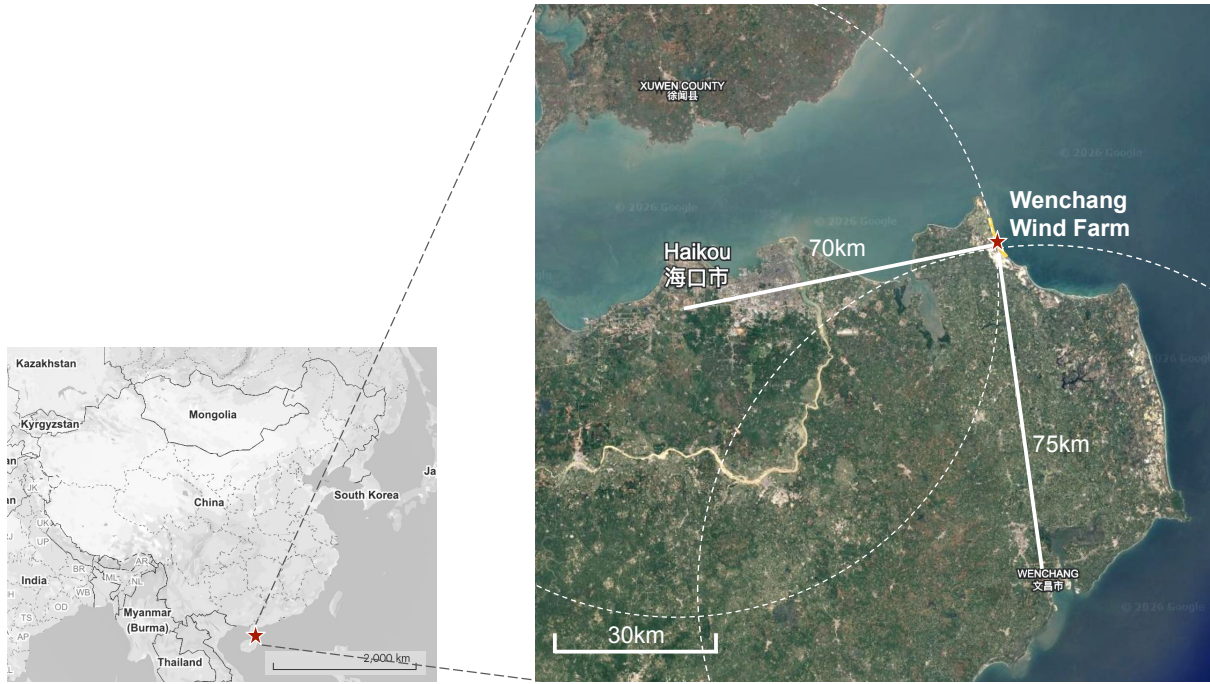


Figure 2: Location of the Wenchang Plant. The wind farm is located on the southeastern coast of Wenchang City in eastern Hainan Province, approximately 70 km from Haikou and 75 km from Wenchang City.

of 22 wind turbine generator units was successfully connected to the power grid. The project was officially commissioned and put into operation on December 28, 2008 (Chen, 2020).

1.1.2 Expansion Phase

The second phase of the Wenchang Plant planned to include 11 wind turbine generator units with a single-unit capacity of 2.0 MW and another 11 units with a single-unit capacity of 2.5 MW, with a total installed capacity of 49.5 MW. The project was officially filed on November 21, 2012. However, it remained unconstructed for an extended period and was later cancelled on October 31, 2018 by the Hainan Provincial Development and Reform Commission in accordance with the National Energy Administration's policies concerning approved but non-commenced wind power projects (China Wind News, 2018).

During this period, a prototype 2 MW direct-drive wind turbine generator unit was installed and put into operation in early 2010 (Huaneng Power International, Inc., 2014). Following the severe damage caused by Typhoon Rammasun in 2014, Huaneng issued a procurement notice in 2015 for two additional 2.0 MW wind turbines (Huaneng Hainan Power Co., Ltd., 2015). Subsequent maintenance and refurbishment tender notices (Huaneng Hainan Power Co., Ltd., 2023b,d) indicate that the installed units during this period included one XE2000/80 wind turbine and two XE2000/96 wind turbine units manufactured by SinoAzure Wind Power Co., Ltd. (SinoAzure). As a result, the total number of wind turbines at the wind farm increased to 34 units. It is therefore inferred that two of the earliest Sinovel turbines were likely replaced during the post-typhoon reconstruction process.

1.1.3 Repowering Project

In June 2023, the National Energy Administration issued the Administrative Measures for the Repowering, Upgrading, and Decommissioning of Wind Farms, which encouraged wind farms that had been grid-connected for more than 15 years or that operated turbine units with capacities smaller than 1.5 MW to undertake repowering and upgrading projects (National Energy Administration, 2023). The policy also stipulated that wind farms reaching their designed service life should be decommissioned unless safety assessments confirmed their continued operational suitability. Under this policy framework, the Wenchang Plant implemented a “replacing smaller turbines with larger ones” repowering and capacity expansion project (Ernst & Young Hua Ming LLP, 2024). On November 14, 2023, the repowering project obtained approval from the Hainan Provincial Development and Reform Commission, including the replacement of smaller turbines with larger-capacity units, the construct a new production area to separating production and living facilities, and the installation of new transformer and transmission facilities. The project planned to replacing 32 aging turbines with 16 new, typhoon-resistant turbines with a rated capacity of 6.25 MW each and increasing the total installed capacity to 104 MW (iMarine Wind Power Network, 2023).

Prior to the upgrade, the wind farm consisted of 34 turbines. Thirty-one Sinovel SL1500/70 wind turbines were commissioned in December 2008 and exited warranty coverage in April 2016, while one SinoAzure XE82-2000 wind turbine unit was commissioned in early 2010 and exited warranty in 2015. In addition, two SinoAzure XE96-2000BF wind turbine units were commissioned in June 2016 and exited warranty in December 2021 (Huaneng Hainan Power Co., Ltd., 2023a; iMarine Wind Power Network, 2023). Based on the commissioning timeline and available maintenance records, it is inferred that the retained turbines, #27 and #33, corresponded to the newer SinoAzure XE96-2000BF units installed after Typhoon Rammasun in 2014, likely as replacements for damaged older turbines. Because these units were significantly newer than the rest of the wind farm fleet, they were excluded from the repowering plan.

Major participants in the Wenchang Wind Farm repowering project included several engineering, consulting, and maintenance contractors. While detailed information regarding the design institute was not publicly available, the feasibility study preparation service and the review of the preliminary design report were awarded to China Huaneng Group Clean Energy Research Institute Co., Ltd. (Huaneng Hainan Power Co., Ltd., 2023f,g). The whole-process cost consulting service contract was awarded to Qingju Engineering Consulting Co., Ltd. for RMB 80,000 (approximately USD 11,000) (Huaneng Hainan Power Co., Ltd., 2023h). The company demonstrated five valid project references during qualification review. Its services covered the full project lifecycle, from feasibility study and preliminary design review through dismantling, civil works, installation, commissioning, and final acceptance. In particular, the consultant was responsible for engineering quantity verification, material quantity inspection and comparison, settlement review, and dynamic cost control throughout construction (Huaneng Hainan Power Co., Ltd., 2023c).

For the main construction including turbine dismantling and installation works, the first-ranked bidder was China Energy Engineering Group Hunan Thermal Power Construction Co., Ltd. (Hunan Thermal Power) with a bid price of RMB 55.90 million (approximately USD 7.7 million), lower than the other shortlisted candidates, China Energy Engineering Group Northeast No.1 Electric Power Construction Co., Ltd. with RMB 64.96 million (approximately USD 9.0 million) and Sinohydro Bureau 5 Co., Ltd. with RMB 56.19 million (approximately USD 7.8 million) (Huaneng Hainan Power Co., Ltd., 2024). The services from Hunan Thermal Power included engineering quality inspection and testing services for construction materials such as

cement and reinforcing steel (European Offshore Wind, 2024). For the wind turbine supply, the shortlisted bidders were Windey Energy Technology Group Co., Ltd. (Windey), Dongfang Electric Corporation, and Sany Renewable Energy Co., Ltd., with bid prices of RMB 148 million (approximately USD 20.6 million), RMB145 million (approximately USD 20.2 million), and RMB 145 million (approximately USD 20.2 million), respectively. Windey was ultimately selected as the supplier (National Business Daily, 2024).

For operation and maintenance, the 2024 daily maintenance contract for the wind farm's 34 existing turbines was awarded to Ruiyuan Wind Energy Technology Co., Ltd. for RMB 1,27 million (approximately USD 0.175 million) (Huaneng Hainan Power Co., Ltd., 2023e). The company provided five valid operation and maintenance references and held a Class III license for power facility installation, repair, and testing. The contract covered routine maintenance of the existing turbines, including inspections, defect handling, supervisory control and data acquisition systems (SCADA) maintenance, and support for major component replacement. The procurement notice specified that the contract applied only to the existing turbines prior to dismantling under the repowering project and did not cover maintenance during construction or operation of the new turbines (Huaneng Hainan Power Co., Ltd., 2023a).

The Wenchang Wind Farm repowering project was developed under the regulatory framework established by the Interim Measures for the Administration of Wind Power Development and Construction issued by the National Energy Administration in 2011 (National Energy Administration, 2011). These regulations govern wind farm planning, approval, construction supervision, grid connection, operational assessment, and reporting requirements. Developers are required to obtain approval from and provincial energy authorities before initiating preliminary project development work. Provincial-level energy authorities are responsible for supervising project completion acceptance and reviewing commissioning documentation after construction and grid connection are completed. In addition, one year after commercial operation, the national energy authority may organize post-operation assessments of project construction and operational performance. The regulations also require immediate reporting of major incidents, including fires, severe turbine damage, prolonged shutdowns, or large-scale component defects, to both national and provincial energy authorities.

The repowering project included 6.25 MW turbines designed for a 50-year return-period wind speed of 50 m/s, with a mean wind speed of 15 m/s at 100 m hub height and an average turbulence intensity of 0.11, requiring The International Electrotechnical Commission (IEC) Class IC or equivalent Class S standards. In contrast, the original turbines were designed to IEC ClassII standards with a hub height of approximately 75m. Under the IEC61400-1 classification system (see Figure 3), Class I turbines are designed for high-wind sites, while the subclass "C" indicates low turbulence intensity conditions.

Wind turbine class		I	II	III	S
V _{ref} [m/s]		50	42,5	37,5	Values to be specified by the manufacturer
V _{ave} [m/s]		10	8,5	7,5	
Onshore	A + I _{ref} [-]	0,18			
	A I _{ref} [-]	0,16			
	B I _{ref} [-]	0,14			
	C I _{ref} [-]	0,12			

Figure 3: IEC61400-1 wind turbine classification system based on reference wind speed (V_{ref}), average wind speed (V_{ave}), and turbulence intensity (I_{ref}). ClassesI, II, and III correspond to high-, medium-, and low-wind-speed sites, respectively, while subclasses A, B, and C indicate decreasing turbulence intensity levels (image from XenogyRE, 2021).

The repowering project dismantled 32 existing turbines while reusing the foundations at 16 selected turbine locations. To strengthen these foundations for the new larger-capacity turbines, the project added 384 new PHC(T)500B125 prestressed high-strength concrete (PHC) anti-uplift pipe piles with a concrete strength grade of C80 (see Figure 4) (International Energy Network, 2024). According to the project contractor, China Energy Engineering Group Hunan Thermal Power Construction Co., Ltd., construction commenced on March 27, 2024. PHC pipe pile installation was completed on June 21, turbine foundation concreting on July 16, and dismantling of all 32 old turbines on July 4 (European Offshore Wind, 2024).



Figure 4: Reinforcement cage and anchor-bolt assembly construction for the repowered wind turbine foundation at the Wenchang Wind Plant. The upgraded foundation system incorporated additional PHC(T)500B125 prestressed high-strength concrete anti-uplift pipe piles to support the larger 6.25 MW typhoon-resistant turbines (image from European Offshore Wind, 2024).

1.2 Wind Turbine

As shown in Figure.5, the main components of a wind turbine include: (1) the rotor, consisting of the blades and hub, which captures kinetic energy from the wind and converts it into rotational motion; (2) the drivetrain, comprising the shafts, gearbox, brakes, and couplings, which transfers and conditions the mechanical power generated by the rotor for electricity production; (3) the nacelle and main frame, which house and support the key mechanical and electrical components such as the generator, gearbox, and control systems; and (4) the tower, which elevates the system to optimize wind capture. A yaw system mounted between the nacelle and the tower rotates the nacelle horizontally to keep the rotor aligned with the wind direction, thereby maximizing energy capture and reducing asymmetric aerodynamic loads on the turbine structure (Flores-Guerrero et al., 2021).

Three different wind turbine models were deployed across different development stages, including the Sinovel SL1500/70 (1.5 MW) in Phase I, the SinoAzure XE96-2000BF (2.0 MW) in expansion, and the Windey WD190-6250 (6.25 MW) turbines adopted in the repowering phase (iMarine Wind Power Network, 2023; Windey Energy Technology Group, 2024). Hub height

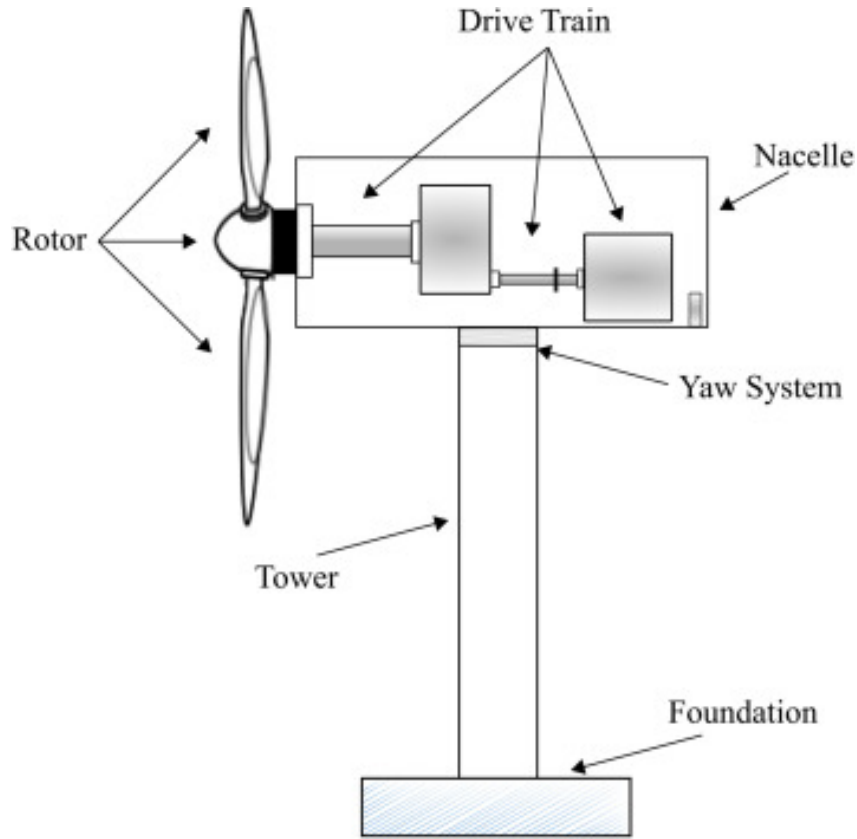


Figure 5: The structure and components of a wind turbine, showing the rotor, nacelle (housing the drivetrain), yaw system, tower, and foundation, with each component labeled according to its function (image from: Flores-Guerrero et al., 2021).

is the distance from the ground to the rotor center and is a critical design parameter influencing turbine performance. Higher hub heights enable turbines to operate in wind regimes with higher mean wind speeds and reduced turbulence intensity due to vertical wind shear effects. In the original wind farm configuration, the turbine hub heights ranged from 75–80m. Under the repowering project, the new replacement turbines adopted a hub height of 108m (Table 1).

The SinoAzure XE96–2000BF turbines installed during the expansion phase of the Wen-chang Wind Plant and retained in the subsequent repowering project were adapted for coastal environments exposed to extreme wind events. According to the manufacturer’s typhoon protection strategy, the turbine adopted enhanced cut-out and shutdown procedures under typhoon conditions, including staged shutdown thresholds based on both 10-minute and 3-second average wind speeds. During grid-connected conditions, the turbine automatically entered a shutdown state and maintained the blades in a feathered position through hydraulic pressure control to reduce aerodynamic loading. In the event of grid loss, an emergency control module enabled the turbine to continue yaw operation using backup battery power, allowing the nacelle to rotate to a safe downwind position while maintaining blade feathering. The yaw control system was also designed with a relatively high yaw speed and wide yaw angle tolerance to respond rapidly to changing typhoon wind directions, thereby reducing asymmetric loading and minimizing the risk of structural damage during extreme wind events (Li and He, 2018).

The Windey WD190–6250 turbines planned for installation during the repowering project incorporated several typhoon-resilient design features intended to improve structural reliability and operational stability under extreme wind conditions. According to the manufacturer, these features included redundant structural design, modular platform-based architecture, inde-

Table 1: Key Specifications of Wind Turbine Models Used in the Project.

Parameter	SL1500/70	XE96-2000BF	WD190-6250
Rated Power (MW)	1.5	2.0	6.25
Cut-in Wind Speed (m/s)	3	2	3
Cut-out Wind Speed (m/s)	25	23	20-25
Survival Wind Speed (m/s)	59.5	60	59.5
Rotor Diameter (m)	77	96	190
Hub Height (m)	75	80	108
Blade Number	3	3	3
Pitch Control	Motor + Gearbox	Variable Pitch	Doubly-fed
Yaw System	Active	Active	Active
IEC Class	IEC II	IEC IA	IEC S

Note: Data compiled from manufacturer websites: Sinovel SL1500 (<http://www.sinovel.com/english/content/?106.html>); SinoAzure XE-series (<http://www.hnxfld.com.cn/?clean/>); and Windey WD175-6000/6250 (https://windeyenergy.com/product_show?id=24).

pendent pitch control technology to reduce fatigue loading, site-specific load optimization to mitigate vibration, and dynamic thrust control strategies for soft-tower resonance suppression. The turbines were also equipped with integrated intelligent monitoring and early-warning systems for real-time operational supervision and fault detection. This turbine model was marketed as “typhoon-resistant” wind turbines, emphasizing their capability to withstand extreme coastal wind conditions through reinforced structural design, reduced aerodynamic exposure, and intelligent typhoon control strategies. However, during the development and construction stages of the projects, there was no unified national technical standard specifically defining the performance requirements or certification criteria for “typhoon-resistant” wind turbines in China.

2. FAILURE OVERVIEW

A decade apart, two super typhoons of comparable intensity struck the Wenchang Wind Plant. The more severe Typhoon Rammasun (2014) caused the failure of two turbines, whereas the slightly weaker Typhoon Yagi (2024) led to the collapse of eight modern large-capacity turbines.

2.1 Typhoon Rammasun (2014)

In July 2014, Super Typhoon Rammasun struck Hainan, Guangdong, and Guangxi, causing severe damage in Wenchang and parts of western Guangdong. The typhoon formed over the northwestern Pacific Ocean on July 12 and made landfall at approximately 15:30 on July 18 in Wengtian Town, Wenchang City, Hainan Province (Figure 6(b)). At landfall, it recorded a maximum sustained wind speed of 60 m/s and a central pressure of 910 hPa, both reaching historical extremes (Figure 6(a)) (Huang et al., 2021; Nanhai Net, 2014). At around 17:10, the typhoon passed near the Wenchang Plant, which at the time 33 turbines of 1.5 MW and a single turbine of 2.0 MW unit (Figure 6(c)) was operating. The storm persisted for four to five hours, causing structural damage to three turbines. Turbine #33 collapsed, while turbines #7 and #27 experienced blade loss (Figure 6(d)) (China Wind News, 2018).

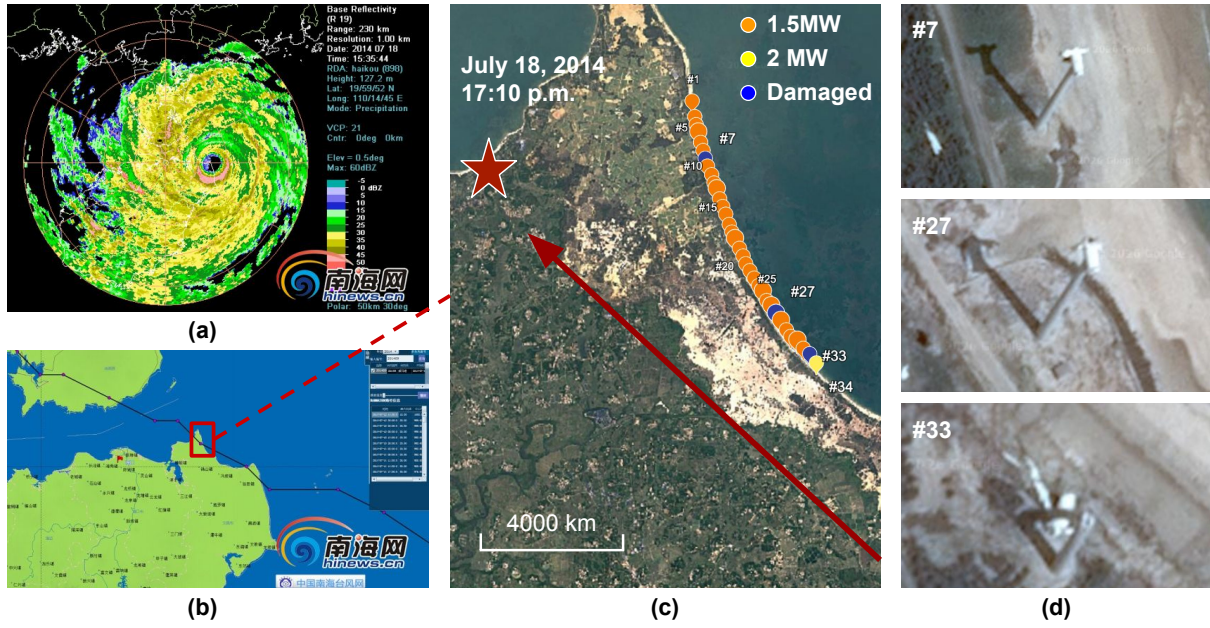


Figure 6: Overview of the Typhoon Rammasun event. (a) Doppler radar image of Haikou; (b) track of Typhoon Rammasun; (c) layout of the Huaneng Wenchang Wind Farm during the event, where orange dots denote 1.5 MW turbines installed in Phase I, yellow dots denote 2.0 MW turbines installed in Phase II, and blue dots indicate turbines damaged by the typhoon; (d) satellite image dated September 25, 2014, showing that turbine #33 collapsed entirely, while the blades of turbines #7 and #27 were torn off (panels (a) and (b) from Nanhai Net, 2014).

2.2 Typhoon Yagi (2024)

In response to Typhoon Yagi, satellite monitoring began on September 3, and the Hainan Meteorological Bureau issued a Level II typhoon warning, indicating a high-intensity landfall (Xinhua News Agency, 2024). On September 4, the project team implemented pre-typhoon mitigation measures (Huaneng Hainan Power, 2024), including foundation drainage inspections, yaw and pitch system checks with manufacturer support, and removal of loose materials along access roads to reduce windborne hazards.

At 16:20 on September 6, 2024, Typhoon Yagi made landfall in Wengtian Town, Wenchang City, passing within approximately 7 km of the project site. As a super typhoon, it recorded a maximum sustained wind speed of 63.9 m/s and a central pressure of 915 hPa (He et al., 2025), exceeding the 60 m/s design limit specified for typhoon-resistant wind turbine systems. Figure 7 illustrates the typhoon track. Yagi was the strongest typhoon to strike China since Rammasun in 2014, affecting over 526,000 people in Hainan Province and causing four fatalities and 95 injuries (Xinhua, 2024).



Figure 7: Track Map of Typhoon Yagi (image from CECS Wind Disaster Mitigation and Wind Energy Utilization, 2024).

At the time, the repowering project was progressing ahead of schedule. Although completion had been planned for the end of October, 15 of the planned 16 turbines had already been installed by early September (Figure 8(a)). These included thirteen newly installed 6.25MW turbines and two retained 2.0MW SinoAzure units (WT-13 and WT-15) (CECS Wind Disaster Mitigation and Wind Energy Utilization, 2024). The numbering WT-1 to WT-15 was assigned by the investigation team, while WT-13 and WT-15 corresponded to the owner's original turbine numbering, namely turbines #27 and #33, respectively.

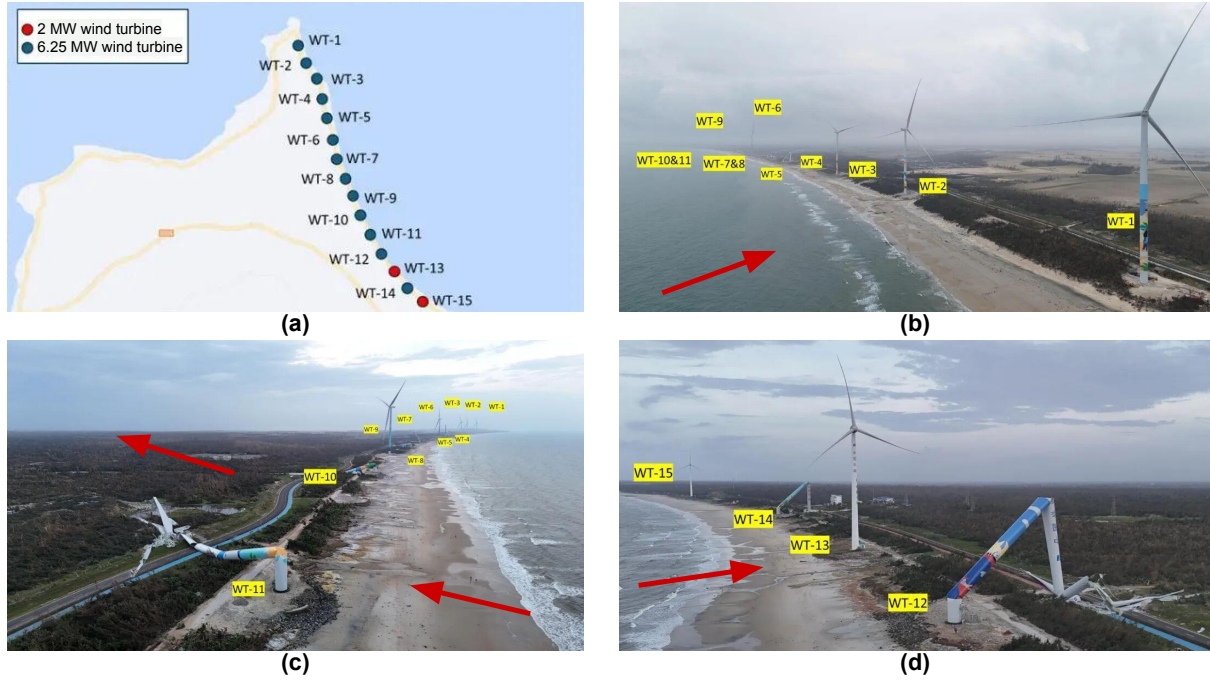


Figure 8: Overview of wind turbine damage induced by Typhoon Yagi. (a) Location of the wind turbines; (b)–(d) post-damage views of the Wenchang Plant from different directions. The red arrow indicates the wind direction (adapted from CECS Wind Disaster Mitigation and Wind Energy Utilization, 2024).

As the project remained under construction, the yaw systems of several turbines were not yet energized and therefore could not align with the incoming wind. This prevented load redistribution through yaw control, resulting in unfavorable structural loading and ultimately leading to tower buckling and collapse.

Figures 8 (b)–(d) present the disaster site from different viewing directions. Typhoon Yagi caused the collapse of eight 6.25 MW turbines (WT-4, WT-5, WT-7, WT-8, WT-10, WT-11, WT-12, and WT-14), while the remaining seven turbines showed no significant structural damage. Notably, all collapsed units were the 6.25 MW turbines, whereas the two 2.0 MW turbines remained intact. The primary failure mode was tower buckling, with no evidence of blade fracture or foundation instability. The turbines collapsed landward, consistent with wind direction and the orientation of fallen trees, typically exhibiting a three-segment fracture pattern (CECS Wind Disaster Mitigation and Wind Energy Utilization, 2024). This failure sequence generally began with local buckling in the upper-middle portion of the tower. After the initial instability occurred, the tower top underwent large horizontal displacement while the buckled section did not immediately separate. The resulting eccentric gravity load from the nacelle and rotor generated significant second-order bending moments at the tower base, leading to a secondary local buckling failure near the bottom of the tower. The tower ultimately fractured into three major segments during collapse (see Figure 9).

Damage details of WT-4, WT-5, WT-7, and WT-14 shown in Figure 10 follow this failure pattern. Severe wrinkling deformation, referring to localized outward and inward folding of the tower shell caused by compressive instability of the steel wall, was observed at fracture locations. In several cases, these localized buckling zones progressed into partial or complete segment separation. The observations suggest that local shell buckling initiated the loss of structural stiffness and load-carrying capacity, which subsequently propagated into overall tower collapse under extreme typhoon loading.

Beyond physical damage, the event had significant financial implications not only for the

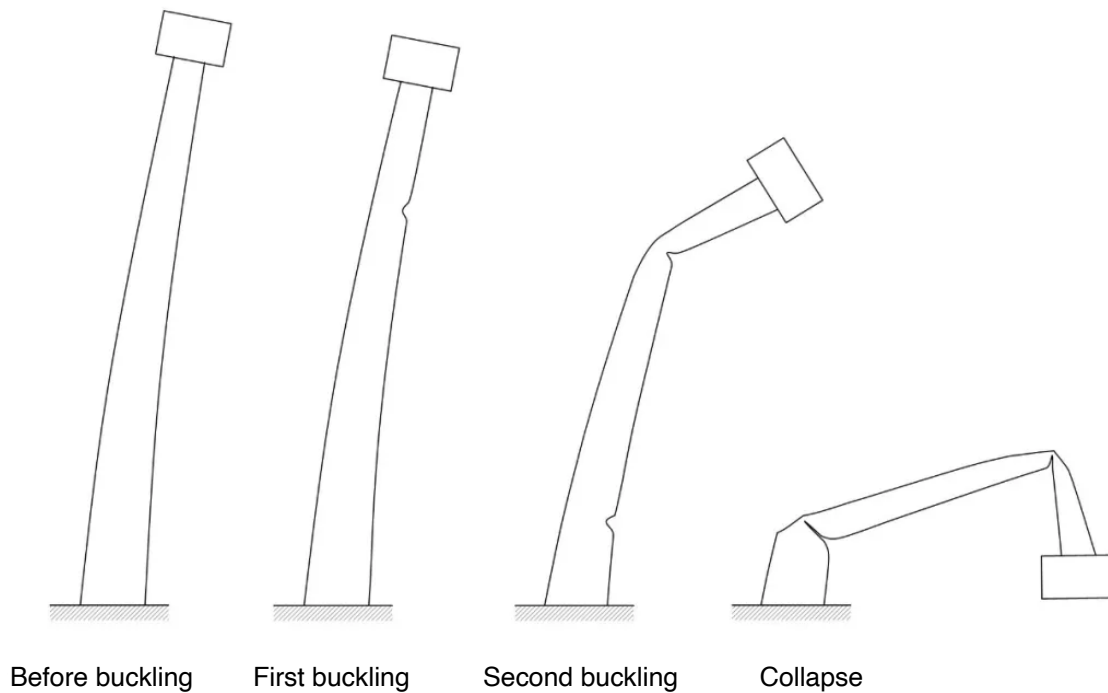


Figure 9: Evolution process of the three-segment fracture pattern of wind turbine collapse under typhoon loading (adapt from from: CECS Wind Disaster Mitigation and Wind Energy Utilization, 2024).

project but also for the associated insurance framework. Each wind turbine has a capital cost exceeding ten million CNY. The assets were insured by Yongcheng Property Insurance Co., Ltd., a power-sector insurer established by major state-owned energy groups including China Huaneng Group. By 2024, Huaneng-related premiums accounted for 30.87% of Yongcheng's total income (Manager Media Center, 2025). Claims related to Yagi reached approximately 940 million CNY, representing 18.9% of Yongcheng's annual payouts (Manager Media Center, 2025).

2.3 After Event

Following the typhoon-induced collapse, emergency response and recovery activities were initiated at the Wenchang Wind Plant. According to subsequent procurement and tender announcements (China Huaneng Group, 2024a,b,c,d), emergency dismantling and cleanup works for the collapsed turbine blades, nacelles, and hubs were initiated on September 26, 2024, including on-site segmentation and disposal of damaged blades. Later post-disaster engineering activities included foundation settlement monitoring services for turbine structures and booster station equipment in December 2024. Replacement tower sections were later custom-fabricated by specialized tower manufacturers rather than procured as part of complete turbine packages. Reconstruction and installation works continued through 2025, and the repowering project was reported to have completed construction and entered operation by May 31, 2025 (Huaneng Hainan Clean Energy Voice, 2025).

The event also contributed to subsequent revisions and development of typhoon-related wind

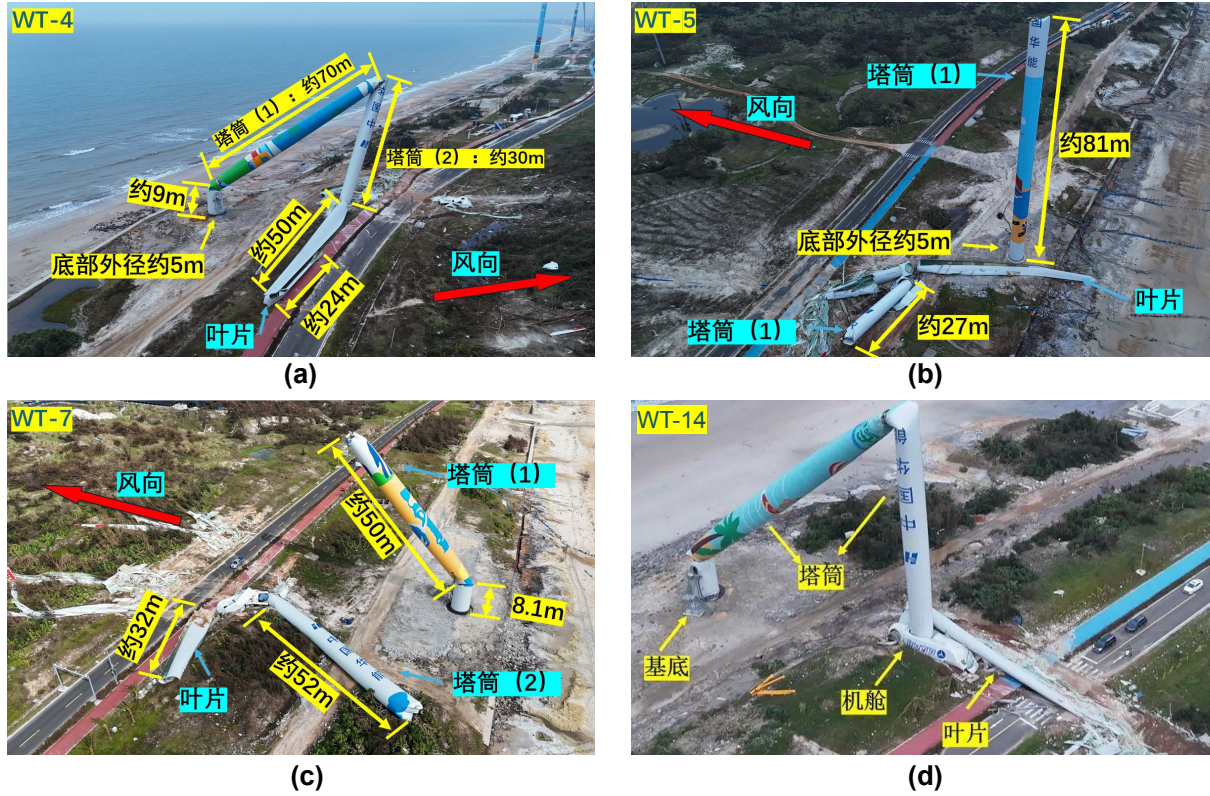


Figure 10: Damage details of turbines WT-4, WT-5, WT-7, and WT-14. The figure illustrates the failure configurations of selected turbines, with major components and estimated dimensions annotated. The red arrows indicate the wind direction during the typhoon event (adapted from CECS Wind Disaster Mitigation and Wind Energy Utilization, 2024).

turbine standards in China. On February 2, 2026, the national standard GB/T 31519–2026, *Wind Energy Generation Systems – Typhoon-Type Wind Turbines*, revised under the leadership of Windey, was officially released. The standards were intended to strengthen typhoon-resistant design requirements and environmental monitoring capabilities for wind power systems in China (Wind.in-en.com, 2026).

3. IDENTIFY CONTRIBUTING FACTORS

The failure of the Wenchang Plant during Typhoon Yagi resulted from the interaction of multiple factors, categorized into seven groups: (1) design, structural redundancy, and cascading effects; (2) codes, materials, and weather; (3) safety and training; (4) contractual and documentation issues; (5) construction, quality control, and assurance; (6) operations, maintenance, and inspection; and (7) financial and other factors.

3.1 Design and Structural Redundancy & Cascading Effects

Underestimation of extreme typhoon loads in turbine selection and design (Responsible party: Huaneng (owner and design reviewer)) The wind turbines installed in the the Wenchang Plant is designed for a 50-year return-period wind speed of 50 m/s, corresponding to IEC Class IC or equivalent Class S standards. This classification is intended to account for site-specific extreme conditions in typhoon-prone regions. However, historical records show that typhoon wind speeds in the region can exceed 60 m/s (Huang et al., 2021; Nanhai Net, 2014;

Wang et al., 2021b), which is higher than the design basis adopted for this project. This suggests that turbine selection may not have fully accounted for known worst-case environmental conditions. Consequently, reliance on nominal design parameters without sufficient adjustment for site-specific extremes may have led to underestimation of actual structural demand.

Floating platforms and approximately 1,700 offshore wind turbines deployed in the affected region remained operational and withstood wind speeds of similar magnitude during the same typhoon event (China Fortune News, 2024). This demonstrates that not all turbines failed under identical environmental conditions, indicating that structural performance depended on turbine-specific design characteristics rather than wind intensity alone.

Lack of redundancy in tower structural design (Responsible party: Huaneng (owner), Windy (wind turbine supplier)) The observed failure of the tower through local buckling indicates that the applied loads exceeded the structural capacity of the tower section (CECS Wind Disaster Mitigation and Wind Energy Utilization, 2024). As a slender, thin-walled structure subjected to combined bending and compression, the tower is inherently sensitive to instability, and once local buckling occurs, the affected region rapidly loses load-carrying capacity, leading to progressive collapse.

Furthermore, tower structures generally lack redundancy, relying on a single primary load path without alternative mechanisms to redistribute loads once local failure initiates. As a result, exceedance of design capacity can lead to rapid and disproportionate collapse. This suggests that, while the design may have complied with standard requirements, it did not provide sufficient reserve capacity or robustness to withstand extreme loading conditions beyond the nominal design basis.

3.2 Codes, Materials, & Weather

Typhoon-prone environmental conditions (Responsible party: Huaneng (owner)) The project site is located in one of the highest typhoon-risk zones in China (Wang et al., 2021b), experiencing frequent landfall of intense storms and an average of 8 to 9 typhoons annually (Haikou Meteorological Bureau, 2024). While such locations provide strong wind resources, they also subject structures to repeated extreme loading. This highlights the importance of evaluating site suitability not only in terms of energy potential but also in terms of long-term exposure to severe environmental hazards.

Code not fully account for extreme typhoon conditions (Responsible party: regulatory authorities) Design standards for offshore wind turbines define extreme wind conditions using parameters such as reference wind speed, wind profile, turbulence intensity, and fluctuating wind field models for load simulation. For typhoon-resistant design in the south and southeast China, these standards can be broadly divided into two types: IEC-based standards (e.g., IEC 61400-3, GB/T 31519), which follow a unified design framework with parameter adjustments, and marine or civil engineering-based standards, which draw on offshore structural experience and adopt different approaches to defining wind loads (Wang et al., 2023).

However, studies by Wang et al. (2023) indicate that such code-based design approaches may not fully capture the characteristics of typhoon-induced wind fields. While IEC-based design conditions can provide a safety margin of approximately 60%–90% under peripheral typhoon effects, turbine responses under eyewall conditions can exceed design predictions by over 35%. This suggests that current standards may not fully capture the intensity and variability of extreme typhoon-induced loads.

In this broader turbine-level design context, the wind turbine tower—being the primary load-bearing component—plays a critical role in resisting these extreme wind actions. Although wind turbine towers are typically designed in accordance with code-based return-period wind loads (50-year events) and standard service life assumptions (20–25 years), such designs may primarily satisfy minimum code requirements rather than account for extreme or site-specific worst-case conditions. Under severe typhoon loading, especially with additional effects such as dynamic wind fluctuations and yaw misalignment, the actual structural demand may exceed these nominal design limits.

Extreme typhoon winds (63.9 m/s) The event involved extreme wind speeds of approximately 63.9 m/s as Typhoon Yagi passed within 7 km of the project (He et al., 2025). This significantly exceeded the design basis of 50 m/s specified for the turbines (National Business Daily, 2024), indicating that the environmental loading conditions were beyond those assumed in standard design scenarios. Such extreme winds, particularly during the eyewall phase, are associated with high turbulence intensity and rapid directional changes that can impose severe and highly dynamic loads on structures.

3.3 Safety & Training

Insufficient storm preparedness planning (Responsible party: Huaneng (owner), Windy (wind turbine supplier)) Pre-typhoon mitigation measures were implemented, including inspection of turbine foundation drainage, coordination with manufacturers for yaw and pitch adjustments, and site cleanup to remove loose materials (Huaneng Hainan Power, 2024). These actions indicate that basic hazard awareness and standard safety procedures were in place. However, the preparation measures appear to have been insufficient for the actual risk conditions.

In particular, the turbines were not yet energized, meaning that active control systems such as yaw and pitch could not operate under extreme wind conditions. Wind turbine structures are primarily designed to withstand extreme wind loads when facing the wind directly; however, when yaw control is unavailable and the turbine is misaligned, lateral wind loading significantly increases structural demand and reduces effective load-bearing capacity (Song et al., 2018). Critically, there is no evidence that alternative measures—such as the provision or activation of backup power supply—were implemented to enable these systems during the typhoon event. As a result, turbines were unable to align with wind direction or reduce aerodynamic loads, leading to substantially increased structural loading and a higher likelihood of failure.

3.4 Construction, Quality Control, & Quality Assurance

Accelerated construction schedule increasing exposure risk (Responsible party: Huaneng (owner), Hunan Thermal Power (construction)) Project records indicate that construction progressed at a notably rapid pace, with key milestones achieved within a highly compressed timeframe. According to the tender requirements, construction was scheduled to begin with the first turbine foundation works by April 30, 2024, and reach full completion after 240-hour trial operation by November 30, 2024 (International Energy Network, 2024).

However, actual construction progressed significantly ahead of schedule. The project commenced on March 27, 2024, with all 32 existing turbines dismantled by early July and foundation works largely completed by mid-July. Reports also indicate that construction activities were carried out under highly intensive schedules (Huaneng Hainan Power Generation Co., Ltd., 2024). The project was expected to be completed by September, earlier than the originally planned November timeline.

This accelerated schedule brought partially installed and non-operational turbines into direct exposure during the Typhoon Yagi event. As a result, critical intermediate construction stages coincided with peak typhoon season, increasing the likelihood that incomplete structural systems were subjected to environmental loading beyond their temporary capacity.

Construction scheduling during typhoon season (Responsible party: Huaneng (owner), regulatory authorities) The construction period coincided with the peak typhoon season in the region, spanning from July to October (Jiang et al., 2018). Conducting critical installation stages—particularly when turbines were incomplete and non-operational—during this period inherently increased exposure to severe environmental loads. This suggests that seasonal risk was not fully integrated into construction planning, leading to elevated vulnerability during a known high-risk timeframe.

3.5 Operations, Maintenance & Inspection

Mismatch between turbine design life and project lifecycle planning (Responsible party: regulatory authorities) Wind farms are typically designed for long-term operation, while individual wind turbines have a shorter service life of approximately 20–25 years. This mismatch necessitates repowering or major reconstruction during the project lifecycle, either due to aging equipment or the need to increase generation capacity.

In this case, the repowering project introduced a transitional phase in which the site operated with partially decommissioned and partially constructed systems. Such conditions inherently increase system vulnerability, particularly when critical control functions are unavailable. While repowering is a common and necessary practice, the timing and management of this transition did not fully account for exposure to extreme environmental conditions. This suggests that lifecycle planning decisions, although economically and operationally justified, contributed to the creation of high-risk conditions during the construction phase, which ultimately increased the likelihood of failure under extreme typhoon loading.

3.6 Contractual & Documentation

Compressed Engineering Survey and Design Schedule (Responsible party: Huaneng (owner)) On June 25, 2023, China Huaneng published the tender announcement for the survey and design services of the Wenchang Plant repowering project. The announcement specifies a highly compressed timeline for investigation and design activities, requiring the contractor to complete the engineering survey within 10 days, submit the preliminary design within 15 days, and finalize the initial design within 30 days of contract initiation (iMarine Wind Power Network, 2023). The timeline is substantially shorter than both typical onshore projects and comparable regional developments. For example, the Longyao County 50 MW wind power EPC project specifies an investigation period of 30 days and a design period of 90 days (Gangxin Zhijia, 2026), while the Dongfang CZ8 offshore wind project in Hainan allocated approximately 3 months for engineering survey and design tasks following contract signing (iMarine Wind Power Network, 2022). A potential reason is that as a repowering of an existing wind farm, the investigation process relied partially on previous site data and assumed that extensive re-assessment was unnecessary.

However, changes in turbine size, layout, and loading conditions associated with larger-capacity turbines require updated and site-specific investigation. The observation that not all turbines collapsed during the event suggests that local site conditions and turbine locations influenced structural performance. According to the National Energy Administration of China (2022), wind turbine micro-siting requires detailed geological assessment, terrain analysis, and

site-specific evaluation during the investigation phase. However, the 10-day investigation period in this project may have been insufficient to perform such analysis, potentially limiting the accuracy of turbine placement and site-specific load assessment.

Lack of formal documentation and integration of previous failure events (Responsible party: Huaneng (owner)) The tender documentation requires a post-project evaluation covering wind resource assessment, turbine siting, equipment selection, and structural design (iMarine Wind Power Network, 2023), but does not explicitly mandate the documentation or integration of lessons from disaster events such as typhoon-induced failures. This is particularly relevant for the Wenchang Plant, which experienced turbine damage during Typhoon Ramma-sun (2014), yet publicly available failure analyses are limited. The absence of such requirements suggests that prior failure events may not have been systematically incorporated into the design process, indicating a gap in the feedback mechanism between past failures and current engineering decisions.

3.7 Financial & Other

Cost-driven procurement and low-price competition (Responsible party: regulatory authorities) The wind power industry has experienced significant price reductions in recent years, with turbine bid prices approaching or even falling below estimated production costs. For example, the weighted average bid price for onshore wind turbines in China has decreased from approximately 4000 RMB/kW in 2012 to around 1400–1500 RMB/kW in recent years, which is close to or even below the estimated manufacturing cost of approximately 1400 RMB/kW (Yujian Energy, 2024). A similar trend is observed in this project, where the winning bid for 6.25 MW turbines was approximately 9.25 million RMB per unit (excluding tower), corresponding to about 1480 RMB/kW (Global Zero Carbon Research Center, 2024). This indicates limited pricing margins and reflects the broader context of intensified price competition within the industry.

Under such conditions, cost-driven procurement practices may incentivize developers to prioritize lower upfront costs over long-term reliability and performance. This can result in turbine designs with reduced safety margins or insufficient adaptation to site-specific environmental conditions. Observations from the event show that newly installed 6.25 MW turbines experienced failure, while some older, smaller-capacity turbines in the same region remained intact. Although structural performance is influenced by multiple factors, this contrast suggests that turbine selection—potentially shaped by cost considerations—may not have fully accounted for robustness under extreme environmental loading.

Concentration of financial risk within affiliated insurance structures (Responsible party: Huaneng (owner)) The financial impact of the event highlights the role of insurance in risk transfer within the power sector. Yongcheng Insurance, which has strong ownership and business ties with major state-owned power enterprises such as China Huaneng, bears a significant share of losses associated with wind power infrastructure damage. Following Typhoon Yagi, compensation related to damage at the Wenchang Plant alone reached approximately 940 million RMB, accounting for about 18.9% of the insurer's total annual payouts (Manager Media Center, 2025).

Despite reporting a net profit of approximately 66 million RMB in the first half of 2025, the company had previously recorded a net loss of 469 million RMB in 2024, largely due to extreme weather-related claims (Manager Media Center, 2025). This contrast illustrates the substantial

financial volatility associated with concentrated exposure to power-sector risks. There is no publicly available evidence indicating that reinsurance mechanisms were applied to significantly redistribute these losses beyond the primary insurer. As a result, financial exposure appears to remain concentrated within closely affiliated entities. This concentration of risk may reduce incentives for conservative design or risk mitigation and amplify the overall economic impact of extreme events.

Schedule pressure due to project restructuring and capacity demand (Responsible party: Huaneng (owner)) The cancellation of Phase II may have increased pressure to expand capacity within the existing project scope, particularly as the original turbines were approaching the end of their service life. This likely created an urgency to accelerate the repowering process to maintain or increase power generation. Such schedule pressure may have influenced project planning and execution, contributing to shortened timelines and increased exposure to risk during construction and transitional phases.

4. DISCUSSION

4.1 Failure analysis

The failure of the Wenchang Plant was not the result of a single deficiency, but rather the combined effect of interacting technical, environmental, and managerial factors that progressively reduced the safety margin between structural demand and capacity, as shown in Figure 11.

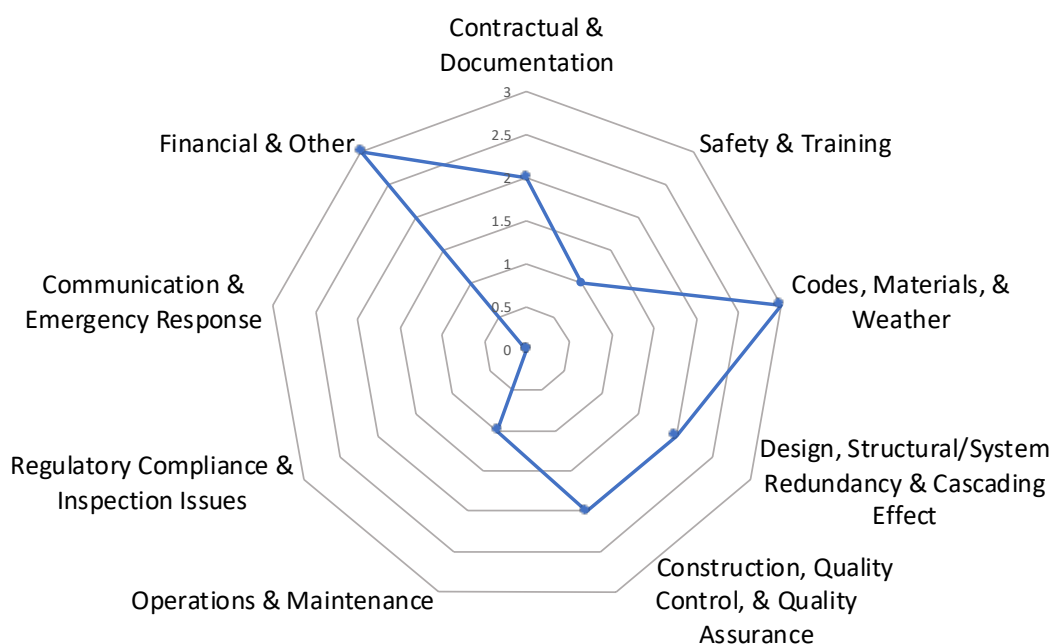


Figure 11: Radar chart of contributing factors

At the design level, extreme typhoon-induced loads were underestimated, as turbine selection and code-based design conditions did not fully account for site-specific wind extremes, particularly under eyewall conditions. This was compounded by the inherent lack of redundancy in tower structures, where local buckling led to rapid loss of load-carrying capacity and progressive collapse once design limits were exceeded.

These design limitations were further amplified by environmental conditions, including the project's location in a high-frequency typhoon zone and the occurrence of wind speeds (63.9 m/s) significantly exceeding the design basis. At the same time, current design standards were not fully capable of representing the intensity and variability of typhoon wind fields, leading to discrepancies between predicted and actual loading. Beyond design, construction-phase vulnerabilities played a critical role. Accelerated scheduling and construction during peak typhoon season exposed partially installed and non-operational turbines to extreme loading. The absence of active control systems (e.g., yaw and pitch), combined with insufficient backup measures, further increased structural demand due to wind misalignment.

Upstream decisions, including compressed survey and design timelines and lack of systematic integration of past failure lessons, likely limited site-specific risk assessment and design adaptation. Additionally, cost-driven procurement and schedule pressure may have reduced design robustness and constrained safety margins. Together, these factors created a condition in which multiple layers of defense—design adequacy, structural robustness, operational preparedness, and planning—were simultaneously weakened, allowing extreme environmental loading to trigger widespread structural failure.

4.2 Ethics

The event emerges from a cumulative chain of technical, environmental, and managerial factors developing over time, as shown in the timeline (see Figure 12). Responsibility for the failure is distributed across multiple stakeholders, each contributing to different aspects of the risk accumulation process.

China Huaneng Group (Owner / Developer) As the project owner, China Huaneng Group held primary responsibility for overall project planning, procurement strategy, and risk management. Decisions such as compressed design timelines, accelerated construction schedules, and cost-driven procurement were made at this level. Given the well-documented history of typhoon impacts in Hainan and previous turbine failures (e.g., Typhoon Rammasun), the owner should reasonably have anticipated extreme environmental risks and ensured more conservative design and scheduling strategies.

Windey Co., Ltd. (Turbine Supplier) As the turbine manufacturer, Windey was responsible for turbine design, including load assumptions and structural capacity. While the turbines may have complied with existing standards, evidence suggests that these standards may not fully capture typhoon conditions. Given increasing industry awareness of typhoon-induced loading uncertainties, the manufacturer could have incorporated more conservative or site-specific design adaptations, particularly for deployment in one of China's highest-risk typhoon regions.

China Energy Engineering Group Hunan Electric Power Design Institute Co., Ltd. (Construction Contractor) CEEC Hunan Thermal Power was responsible for engineering design, site investigation, and construction execution. The compressed investigation period (10 days) and accelerated construction schedule suggest limitations in site-specific assessment and risk integration. Additionally, construction planning during peak typhoon season exposed incomplete structures to extreme hazards. As a professional engineering contractor, they should have identified these risks and recommended adjustments, even under schedule pressure.

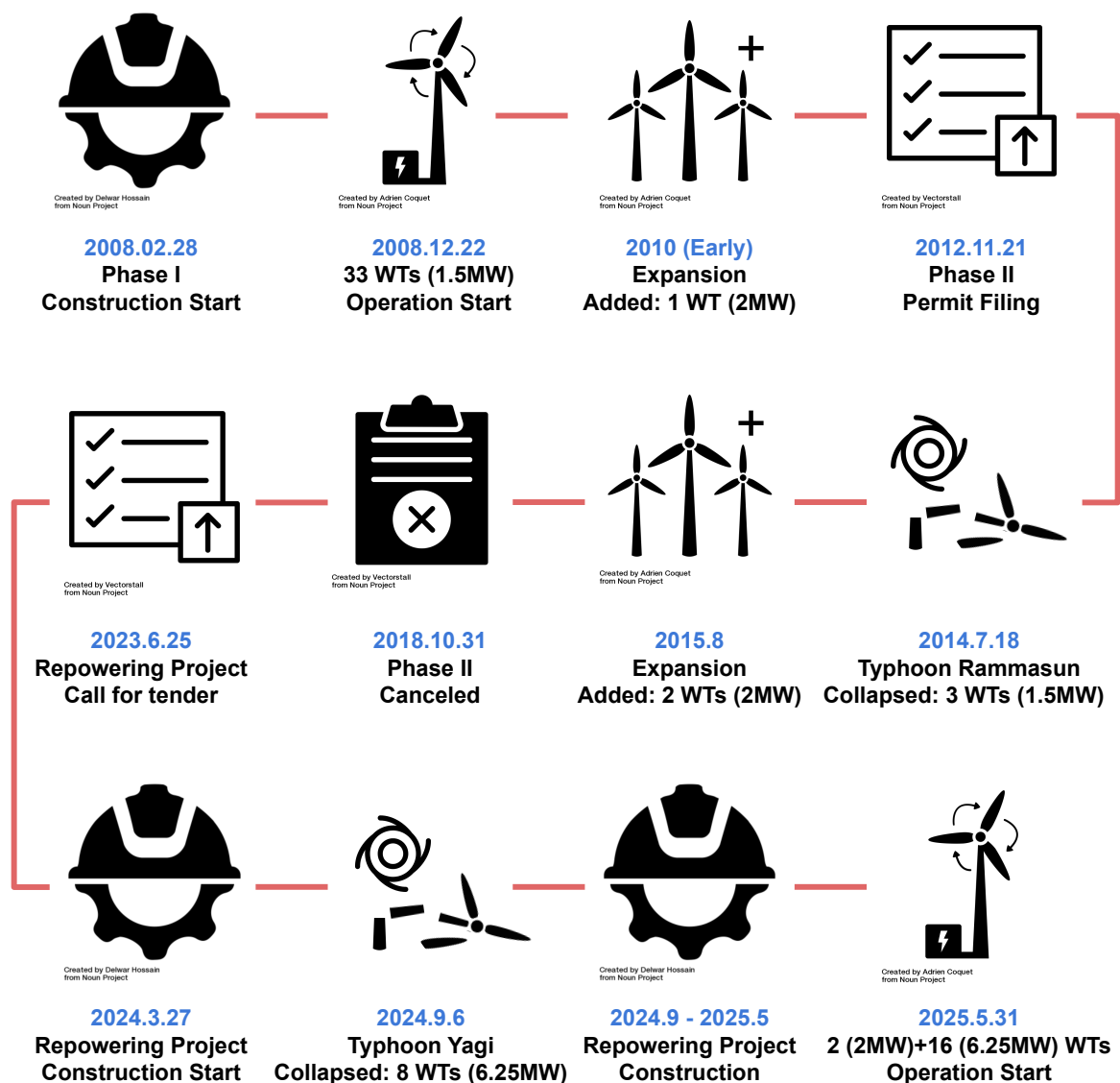


Figure 12: Timeline of the offshore wind project, showing key phases, expansions, disruptions due to typhoons, and the repowering process from initial construction in 2008 to planned operation in 2025.

Regulatory Authorities Including the Hainan Provincial Development and Reform Commission, the National Energy Administration, and the Southern Energy Regulatory Bureau Regulators were responsible for project approval, oversight, and enforcement of safety standards (General Department of the National Energy Administration, 2023). While the project complied with existing codes, the event highlights limitations in those standards for typhoon-prone regions. Given prior regional experience with typhoon damage, regulators could have imposed stricter requirements for site-specific extreme wind assessment, construction scheduling constraints, and resilience measures for non-operational turbines.

5. TECHNOLOGY OPPORTUNITIES

5.1 Insufficient Design Resilience: Passive-Aligning and Bladeless Wind Turbine Designs

The event highlights the need to enhance the intrinsic resilience of wind turbine systems. The observed failures were closely associated with limitations in tower structural design (Section 3.1), particularly the lack of redundancy and sensitivity to extreme loading. This indicates that resilience must be addressed not only through operational strategies but also through improvements in structural and aerodynamic design.

Systems like the Mingyang Smart Energy OceanX platform (Figure 13) use floating foundations with single-point mooring. This innovative “double-head” single-point mooring design reduces the ultimate load on the support structure by approximately 40% and enables the wind turbine to yaw adaptively through the combined effects of anchor-chain traction and the aerodynamic lift generated by the airfoil-shaped tower (Qin et al., 2025).



Figure 13: OceanX floating wind turbine system with dual-rotor configuration. The platform integrates two counter-rotating turbines on a single floating foundation, combining enhanced energy capture with improved structural stability under extreme loading conditions (image from BBC, 2025).

Meanwhile, passive-aligning floating wind turbines, such as vortex-induced vibration bladeless turbines developed by Vortex Bladeless (see Figure 14), generate energy through oscillations induced by vortex shedding rather than rotating blades, reducing mechanical complexity and structural stress under extreme conditions (Mohamed et al., 2025). Unlike conventional turbines, these systems eliminate rotating blades and complex drivetrain components, thereby reducing mechanical complexity and susceptibility to damage. As described in the Vortex Bladeless concept, the absence of gears, bearings, and rotating shafts minimizes wear and potential failure points (Yáñez Villarreal, 2018). Moreover, under very strong wind conditions, the system can naturally exit its resonance state, reducing aerodynamic loading without requiring active braking systems. This behavior contrasts with conventional turbines, where failure can occur when loads exceed design limits, and suggests a potentially higher tolerance to extreme wind conditions.

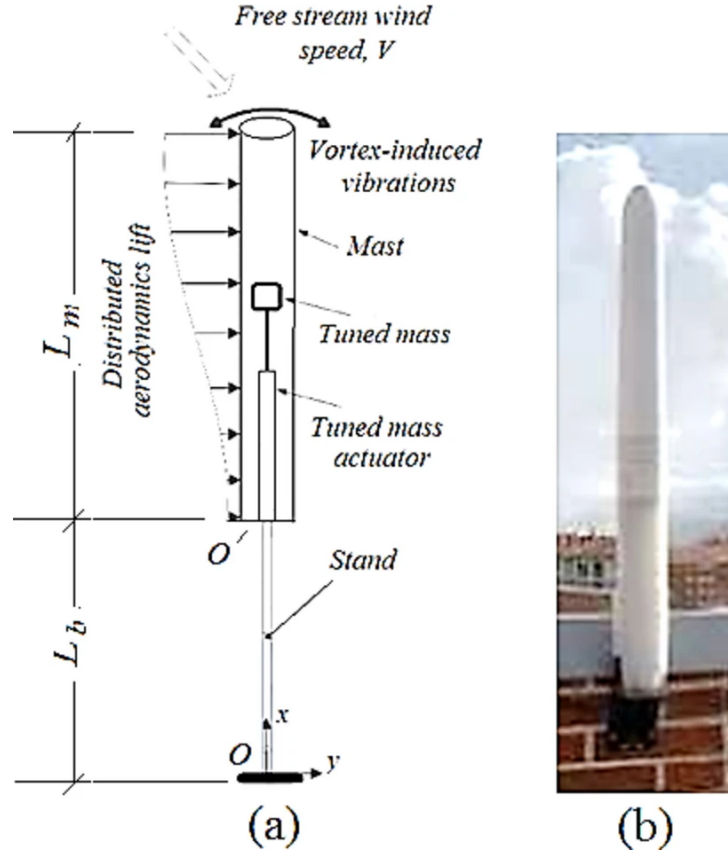


Figure 14: (a) Schematic drawing of a vortex-induced vibration bladeless wind turbine (VIV-BWT). (b) Example of a prototype structure (image from Mohamed et al., 2025).

These technologies are justified because they directly address the primary failure mechanisms identified in extreme wind events. In the Wenchang event, several turbines were unable to align with the wind due to inactive control systems, leading to significantly increased lateral loading. Passive-alignment systems reduce dependence on active control systems, which are often compromised during storms, thereby lowering lateral loads and structural stress on the tower. This enhances safety and reliability while improving return on investment by reducing failure risk and maintenance costs. Bladeless turbines further improve resilience by eliminating critical components such as blades, gearboxes, and rotating shafts, which are common points of failure. Although their current energy efficiency is lower, their reduced maintenance requirements, improved safety, and ability to operate in harsh environments provide long-term economic and operational benefits, particularly in remote or high-risk locations (Mohamed et al., 2025).

Implementation would involve collaboration between turbine manufacturers, offshore developers, and structural engineers to integrate these technologies into future wind energy projects. Passive-aligning floating turbines would be deployed primarily in offshore environments prone to extreme weather, such as typhoon-affected regions, while bladeless systems could be installed in both offshore and onshore settings where maintenance access is limited. Deployment can begin with pilot projects in the near term, followed by gradual scaling over the next decade as performance data and cost efficiencies improve. The process would include replacing fixed-bottom systems with floating platforms, incorporating passive yaw capabilities, and integrating bladeless turbines as complementary systems in hybrid wind farms.

However, these technologies are not without limitations. Floating turbines involve higher initial costs and more complex installation procedures particularly due to long-term testing,

which took approximately five years for the OceanX platform developed by Mingyang Smart Energy (Mingyang Smart Energy, 2024). Bladeless turbines in contrast, currently face lower energy conversion efficiency and limited large-scale validation (Mohamed et al., 2025). Additionally, long-term durability under continuous extreme conditions remains an area requiring further research.

5.2 Inadequate Typhoon Prediction: AI-Enhanced Forecasting

The failure analysis indicates that insufficient preparedness for extreme environmental conditions contributed to the severity of the event. Although general typhoon warnings were issued prior to landfall and standard mitigation measures were implemented (Huaneng Hainan Power, 2024), the project remained in a vulnerable transitional state. In the absence of backup power supply or alternative mitigation measures, turbines were unable to adjust their orientation or reduce aerodynamic loads, directly contributing to the observed structural failures. In this context, earlier and more accurate prediction of typhoon intensity, trajectory, and timing could have supported more effective preparedness and mitigation strategies, including delaying construction activities, securing partially completed structures, or ensuring the temporary activation of critical control functions.

Recent advances in artificial intelligence (AI) provide new opportunities to enhance typhoon forecasting. AI techniques have demonstrated strong capabilities in handling large and complex meteorological datasets (Pradhan et al., 2017), offering improvements over traditional numerical weather prediction methods. In terms of data processing, machine learning algorithms can automatically extract relevant information from diverse data sources, including satellite imagery, radar observations, and ocean measurements, reducing human intervention and improving efficiency (Wang et al., 2021a). In data analysis, deep learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) can identify complex patterns in typhoon formation and evolution, enabling more accurate prediction of storm tracks, intensity, and duration (Giffard-Roisin et al., 2020). Recent developments have further expanded the capabilities of AI-based forecasting systems. A number of large-scale models, including GraphCast, Pangu Weather, and Fengwu, have been developed using advanced architectures such as transformers and graph neural networks (see Table 2). These models integrate atmospheric, oceanic, and land-surface data to produce high-resolution forecasts with reduced computational time compared to traditional numerical models. Their ability to rapidly process large-scale data enables near real-time forecasting, which is particularly valuable for extreme weather events where timely decision-making is critical.

Improved forecast precision enables earlier and more informed decision-making, allowing operators to delay construction, secure partially completed turbines, or ensure backup systems remain active. This reduces structural exposure during vulnerable phases and enhances overall safety. From a cost and return-on-investment perspective, the upfront investment in AI forecasting infrastructure is relatively small compared to the potential financial losses associated with turbine failure, repair, and downtime. Additionally, these models significantly increase forecasting speed, producing results in minutes rather than hours, which is crucial for rapidly evolving storm systems.

Implementation would involve collaboration between meteorological agencies, wind energy developers, and data scientists. Non-profit organizations and private technology developers would provide forecasting models and data infrastructure, while wind farm operators integrate these outputs into their decision-making systems. Deployment would occur primarily in high-risk regions, including offshore wind farms in typhoon-prone areas. In practice, AI forecasts

Table 2: AI weather forecast models for global medium-range scale (data from Bi et al., 2025).

Model	Agency	Resolution	Forecast Days	Layers	Variables
Pangu-Weather	Huawei	25 km, 1 h	7	13	5
GraphCast	DeepMind	25 km, 6 h	10	37	5
FengWu	Shanghai AI Lab	25 km, 6 h	10.75	37	5
AIFS	ECMWF	25 km, 6 h	10	13	10
Fengqing	CMA	25 km, 6 h	15	13	4
FuXi 2.0	Fudan University	25 km, 6 h	15	13	5

would be incorporated into operational protocols, triggering predefined mitigation actions when certain thresholds are met. This could be implemented immediately through pilot programs and scaled over the next several years as model reliability and accessibility improve. When integrated with engineering decision-making processes, improved forecasting can reduce exposure during critical project phases, enhance operational preparedness, and ultimately lower the likelihood of failure under extreme environmental conditions.

The application of AI in typhoon forecasting also faces several challenges. First, model performance is highly dependent on the quality and availability of data. Incomplete or noisy observational data can reduce prediction accuracy, particularly in regions with limited monitoring infrastructure (Chen et al., 2019). Second, AI models often require significant computational resources, which may limit their accessibility for operational use. Third, the interpretability of AI-based predictions remains a concern, as many models operate as “black boxes,” making it difficult to assess the physical consistency of their outputs (Sinha and Lee, 2024). This can affect trust and adoption in engineering decision-making contexts.

5.3 Inadequate Operational and Maintenance Planning: Maintenance and Decision-Making Using Structural Health Monitoring and Partially Observable Markov Decision Process

The failure analysis highlights a fundamental limitation in current maintenance and operational planning approaches, particularly during transitional phases such as repowering. As discussed in Section 1.5, the mismatch between turbine service life and the overall project lifecycle necessitates periodic replacement or upgrading of turbines. In the Wenchang project, this resulted in a temporary state in which partially decommissioned and partially constructed systems co-existed, increasing system vulnerability, especially when critical control systems were not yet operational. Combined with exposure to extreme typhoon conditions, this reveals the limitations of conventional approaches that rely on fixed schedules and deterministic assumptions.

A key enabling technology to address this issue is Structural Health Monitoring (SHM) combined with probabilistic decision-making frameworks such as a Partially Observable Markov Decision Process (POMDP) (Kim et al., 2018). SHM uses sensor-based systems to continuously track structural responses such as vibration, deformation, and loading (Dekker and Scarf, 1998). Unlike periodic inspections that provide only limited snapshots of system condition, SHM offers continuous and site-specific information on structural performance (Farrar and Worden, 2012). However, monitoring alone is not sufficient without a systematic framework for decision-making under uncertainty. A key challenge is that the true structural condition cannot be directly observed and must be inferred from incomplete and uncertain data. To address this, maintenance and operational planning can be formulated within a probabilistic decision-

making framework, such as a POMDP. As illustrated in Figure 15, the POMDP framework represents the evolution of the system over time. The actual structural condition (denoted as the hidden state) cannot be directly observed, but generates measurable signals through inspections or monitoring data. Based on these observations, decisions (e.g., inspection, maintenance, or operational actions) are taken, which in turn influence the future condition of the system. At each step, the decision-maker updates their understanding of the system condition using available information and selects actions accordingly. This framework enables the explicit consideration of uncertainty in both structural behavior and environmental loading. Prior studies (Papakonstantinou and Shinozuka, 2013, 2014) have successfully applied POMDPs to integrate inspection, monitoring, and maintenance actions under uncertainty.

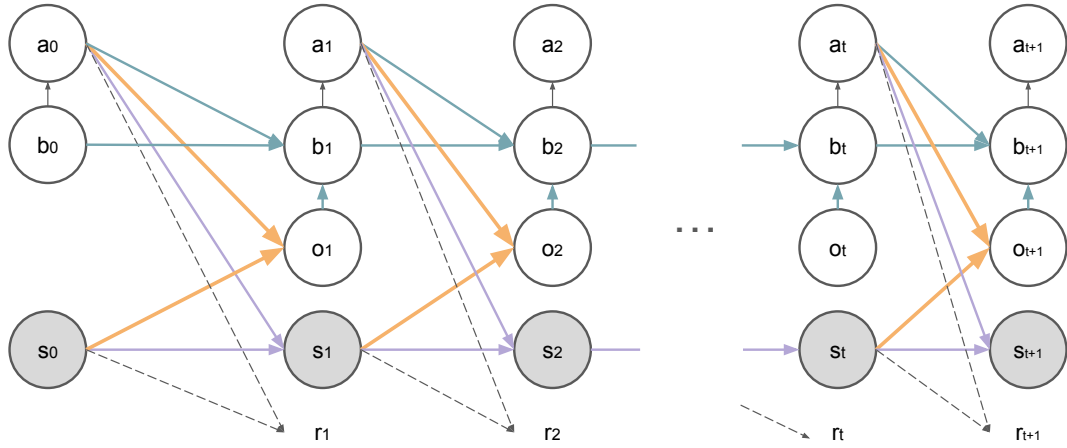


Figure 15: Schematic representation of a Partially Observable Markov Decision Process (POMDP) for infrastructure maintenance. The true structural condition (s_t) evolves over time but is not directly observable. Instead, observations (o_t), obtained from inspections or monitoring systems, provide indirect information about the system state. Based on these observations, actions (a_t), such as inspection, repair, or operational adjustments, are selected. These actions influence future system states and generate rewards or costs (r_t). The framework enables sequential decision-making under uncertainty by continuously updating the estimated system condition and adapting actions accordingly.

The integration of continuous monitoring and probabilistic decision-making directly addresses several issues identified in this event. This is particularly relevant given that not all turbines failed under identical conditions during the event, suggesting that local variability in structural response and site characteristics played an important role. Continuous monitoring enables detection of these differences in real time, while the POMDP framework provides a structured method to act on this information. This improves safety by allowing earlier intervention and reduces the likelihood of catastrophic failure. The approach also supports decision-making under partial system availability. In situations where critical functions such as yaw control are not operational, alternative mitigation strategies can be evaluated based on the estimated level of risk. In addition, it facilitates a shift toward condition-based maintenance, where interventions are triggered by observed or inferred degradation rather than fixed schedules, improving both reliability and resource efficiency. At the same time, this approach provides a structured mechanism for incorporating knowledge from past failure events into current practice. Historical data, including previous typhoon-induced damage, can be integrated into the decision-making process to improve the estimation of system behavior under extreme conditions, addressing the gap in feedback and learning identified in the documentation process.

Implementation would involve collaboration between wind farm operators, structural engi-

neers, and data scientists. SHM systems would be installed on turbines to collect continuous data on structural response, while computational models based on POMDP would be integrated into operational decision-support systems. This approach would be applied primarily in high-risk environments, including offshore wind farms and sites undergoing repowering or construction. Deployment can begin during the design or retrofitting phase and continue throughout the operational lifecycle, with immediate application possible for new projects and phased adoption for existing infrastructure. In practice, sensor data would be continuously collected and processed to estimate the current structural condition. Based on this information, the POMDP framework would update the probability distribution of system states and recommend optimal actions, such as maintenance, inspection, or operational adjustments. This enables adaptive management under uncertainty, particularly in situations where critical systems are not fully operational.

However, this approach also has limitations. Its effectiveness depends on the quality and reliability of sensor data, and the implementation of SHM systems can involve significant initial costs and technical complexity. Additionally, POMDP models require careful calibration and may be computationally demanding, which can limit real-time application in some cases.

6. CONCLUSIONS

This report examined the failure of the Wenchang Wind Power Plant during Typhoon Yagi. The event occurred under extreme environmental conditions, with wind speeds exceeding the design basis. Critically, the severity of the damage cannot be attributed to environmental loading alone, the failure also resulted from the interaction of multiple contributing factors.

At the technical level, the failure was driven by structural and mechanical vulnerabilities. The lack of redundancy in tower systems reduced the ability to withstand extreme loads, while inactive yaw systems increased lateral loading and contributed to tower buckling. At the operational level, risk was elevated by project execution conditions. Turbines were in partially installed or non-operational states, leaving them unable to actively respond to wind conditions. Their exposure during peak typhoon season further increased susceptibility to extreme environmental loading. At the managerial level, upstream decisions played a critical role in shaping these vulnerabilities. Compressed design timelines, limited incorporation of past typhoon-related failure experience, and cost-driven procurement strategies collectively reduced overall safety margins and resilience. The event demonstrates that compliance with existing design standards does not necessarily ensure robustness under extreme and evolving environmental conditions. Responsibility for the failure is distributed across multiple stakeholders, with China Huaneng Group playing a leading role in overall project decisions and risk management. The outcome reflects combined limitations in design assumptions, construction practices, and regulatory frameworks in addressing typhoon-related risks.

Several opportunities emerge for improving the resilience of wind energy systems. Structural design innovations can reduce sensitivity to extreme loading and misalignment. Advances in artificial intelligence for typhoon forecasting provide the potential for earlier and more accurate risk anticipation. In parallel, data-driven maintenance and decision-making frameworks, supported by structural health monitoring and probabilistic models, allow for adaptive management under uncertainty. Beyond technological improvements, the findings also highlight the importance of integrating risk considerations across the entire project lifecycle. This includes aligning construction schedules with environmental risk patterns, incorporating lessons from past failures into design and planning processes, and ensuring that operational contingencies are in place for non-standard system states.

The failure can be considered typical in terms of the number of causes and responsible

parties, as it involved multiple interacting structural, environmental, and operational factors managed by conventional stakeholders. However, it is distinct in the type of cause and timeline: the event occurred during a vulnerable transitional phase when critical systems were not yet operational, and the project had previously survived a stronger typhoon with older technology but failed under less extreme conditions. This contrast highlights that system readiness and project phase can be more decisive than technological advancement alone, emphasizing the importance of decision-making and risk management throughout the project lifecycle, as well as the need for clearer standards of typhoon resilience.

Overall, the Wenchang event illustrates that resilience in wind energy infrastructure is not determined by a single factor, but by the robustness of the entire system—including design, construction, operation, and governance. Addressing these interdependencies is essential for ensuring the safe and reliable deployment of wind power in typhoon-prone regions under increasingly uncertain climate conditions.

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